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SUPERSEDING
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MILITARY SPECIFICATION

COPPER ROD FOR CRUSHER CYLINDERS OR PRESSURE CYLINDERS

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers one grade of electrolytic copper rod in the cold-drawn condition used in the manufacture of crusher cylinders or pressure cylinders for cannon, small arms, pistols, and revolvers.

1.2 Sizes. - Rod shall be furnished in the following sizes, as specified:

0.252 inch diameter
0.2255 inch diameter
0.1456 inch diameter

(For intended use of various sizes, see 6.1.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS,AND PUBLICATIONS

2.1 The following specification and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

FEDERAL

QQ-M-151 - Metals; General Specification for Inspection of

STANDARD

MILITARY

MIL-STD-129 - Marking of Shipments

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.

3.1.1 Material shall be of the best grade of electrolytic copper rod made by either hot rolling or extrusion and finished by cold-drawing to the specified diameter.

3.2 Chemical requirements.

3.2.1 The material shall contain not less than 99.90 percent copper, silver being counted as copper.

3.2.1.1 A chemical analysis of each lot of material shall be furnished by the manufacturer.

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3.3 Dimensions and tolerances.

3.3.1 Diameter. - Rods shall not vary from the ordered diameter, nor be out-of-round, by more than the amounts shown in Table I.

Table I. - Permissible variations in diameter and out-of-roundness.

Ordered diameter	Permissible variations	
	Diameter, plus only	Out-of-round 1/
Inch	Inch	Inch
0.252	0.001	0.001
.2255	.001	.001
.1456	.001	.001

1/ Out-of-round is the difference between the maximum and minimum diameter of the rod measured at the same cross-section.

3.3.2 Length. - Unless otherwise specified, rods shall be furnished in random lengths of 6 to 10 feet.

3.3.3 Straightness. - The copper rods shall be straight within the limits of good commercial practice.

3.4 Tensile requirements. - Rods shall conform to the tensile requirements shown in Table II.

Table II. - Tensile requirements.

Ordered diameter	Tensile strength, minimum	Elongation in 10 inches, minimum
Inch	PSI	Percent
0.252	45,000	1.75
.2255	50,000	1.25
.1456	55,000	0.75

3.5 Electrical Resistivity. - The resistivity of the rod as supplied shall not exceed 0.15940 ohm (meter, gram) at 20°C. (This resistivity is equivalent to 1.7930 microhm-centimeter). The density of the copper shall be taken as 8.89 grams per cubic centimeter (0.32117 pounds per cubic inch) at 20°C.

3.6 Identification. - Each lot shall be identified by a serial number assigned to lots consecutively by the contractor and, following the serial number, by the first letter of the name of the contractor.

3.7 Workmanship. - Rods shall be uniform in quality and temper, clean, sound, smooth, bright, and straight and shall be free from pits, die marks, pipes, slivers, laps, cracks, twists, seams, non-metallic inclusions, damaged ends and other injurious defects. No joints shall be made in the material at any time during fabrication.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Lot. - A lot shall consist of all rods of the same size from the same billet.

4.2 Samples for Government analysis shall be taken from at least 10 percent of the rods in the lot. Each sample shall consist of not less than 2 ounces of drillings or millings and shall be analyzed separately.

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4.3 Tension tests.

4.3.1 At least two tension tests shall be made for each lot.

4.3.2 All rods shall be pulled in full size.

4.3.3 Tensile properties shall be determined by the methods prescribed in Federal Specification QQ-M-151.

4.3.4 Re-tests. - In case of failure to meet the tensile requirements, retests shall be allowed as provided in Federal Specification QQ-M-151.

4.4 Electrical Resistivity Test. - At least 2 samples shall be selected from each lot of rods and tested for electrical resistivity. The rods shall be tested in the hard-drawn condition, shall be full size and at least 1 foot in length. The resistivity determination shall be made with a suitable potentiometer or Kelvin type double bridge method capable of an accuracy of 0.05 percent. Measurements shall be made at, or corrected to, a temperature of 20°C.

4.5 Dimensional inspection. - All rods shall be gaged with an instrument capable of measurement to at least .0001 inch to determine compliance with diameter requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing. -

5.1.1 Segregation. - All material shall be separated by lot when packed for shipment.

5.1.2 All rods shall be packed in boxes or crates so constructed as to insure acceptance for safe transportation by common or other carriers to the point of delivery.

5.2 Marking. - Each shipping container shall be marked with the name of the material, the lot number, the diameter, the weight contained, this specification number, the name of the contractor and the number of the contract or purchase order, and in addition shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - The copper rod covered by this specification is intended for use in the manufacture of copper crusher cylinders or pressure cylinders for determining gas pressures in the testing of propellants and ammunition. Due to the accuracy required in the gas pressure measurements, uniformity in chemical composition and in tensile properties is of the greatest importance. The various sizes of rod are intended for use as follows:

Diameter	Intended for crusher or pressure cylinders for
0.252	Major and medium caliber cannon.
0.2255	Small arms and minor caliber cannon.
0.1456	Pistols and revolvers.

6.2 Ordering data. - Purchasers should specify the title, number, and date of this specification, the diameter of the rod, and should exercize any desired option offered herein (see 3.3.2).

NOTIVE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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